Brij Raj Kishore

Staff Software Engineer · **C++** · **Java** · **Python**
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Professional Summary

Systems software engineer (5.5+ yrs) specializing in **high-performance C++, Java, and Python** for ML and data systems. Currently building a **PyTorch out-of-tree device backend for low-latency LLM inference**, on a foundation of query-engine internals (Velox, Gluten, Spark), **JVM** runtimes, and heterogeneous compute (FPGA, CUDA). **12 merged PRs** across PyTorch, Velox, Spark, and Gluten.

Work Experience

- **ZettaBolt Technologies Pvt. Ltd** — Noida, India
  Senior Member of Technical Staff (Staff Software Engineer) — *December 2021 – Present*
  - **PyTorch OOT Device Backend for LLM Inference**: Building an out-of-tree (OOT) **C++ device backend** in PyTorch (PrivateUse1) – operator dispatch, kernel registration, device memory, streams – against PyTorch's **OpenReg** reference device ahead of custom silicon; validated via the **OpInfo** framework and ran **Llama and BERT inference end-to-end**. Built a **CUDA-parity testing framework** + CI; extending the plugin toward **vLLM serving** (`.to()`, streams, kernels).
  - **Healthcare ETL on Microsoft Fabric & Azure**: **Cut debugging time ~70%** leading optimization of healthcare analytics pipelines on **Apache Spark (PySpark/Scala)**, **Microsoft Fabric Lakehouse**, and **Azure Data Factory**; resolved join skew and AQE fallback via profiling and code review.
  - **Spark Query Engine Acceleration via FPGA Offloading**: Achieved **5x speedup over vanilla Spark** and **2x over Apache Gluten** by integrating **Velox (C++)** with FPGA SDK through a **Scala** bridge layer; designed **scalable** operator dispatch architecture for Scan, Filter, and Aggregation.
  - **AMD AOCL Integration on Meta Velox**: Delivered **5x runtime improvement** on AMD EPYC servers by integrating AMD Compiler and AOCL into Meta's Velox/Gluten pipeline using **C++ and Scala**; guided by **profiling** with uProf and Flamegraph.
  - **JVM-to-C++ Hive Execution Offload**: Improved Hive query runtime by **20%** by migrating execution kernels from the **JVM/Java layer to a C++ backend**, enabling heterogeneous CPU/FPGA dispatch with improved **low-latency** response.
  - **ZettaProf – Automated Spark Profiling & Benchmarking Tool**: **Reduced Spark debugging time 60%** by designing and shipping **ZettaProf** (Scala/Java), an automated profiler adopted **company-wide**; mentored engineers on its use.
  - **Graph Analytics: NVIDIA cuGraph & TigerGraph**: Achieved **100x speedup** on PageRank using TigerGraph GSQL with **NVIDIA CUDA (C++)** and Python; cut pipeline runtime by **40%** and infra cost by **24%**.
- **NXP Semiconductors** — Pune, India
  Software Engineer — *July 2021 – December 2021*
  Ported BT firmware to dual-antenna platform in **C**; debugged WiFi/BLE driver bugs via GDB; validated via **unit tests**.
- **Tata Consultancy Services** — Bengaluru, India
  Assistant System Engineer — *Aug 2017 – Jan 2018*
  Developed **REST API** endpoints (TCS OmniStore POS); raised unit test coverage **62% to 70%** via JUnit and **CI/CD**.

Education

- **National Institute of Technology (NIT Nagpur)** — Nagpur, India
  M.Tech – Computer Science & Engineering; CGPA: 8.04 / 10 — *July 2019 – June 2021*
- **GL Bajaj Institute of Technology & Management** — Greater Noida, India
  B.Tech – Computer Science & Engineering — *July 2013 – June 2017*

Technical Skills

- **Languages**: C++17, C, **Java**, Scala, Python, Bash
- **ML / AI**: PyTorch (OOT device backend, PrivateUse1, OpenReg, operator dispatch, kernel registration, OpInfo, streams), CUDA-parity testing, NVIDIA CUDA, cuGraph, vLLM
- **Query Engines**: Meta Velox, Apache Gluten, Apache Hive, Spark SQL, Parquet/ORC I/O
- **Hardware**: FPGA (Xilinx/AMD), AMD EPYC / AOCL, GDB, Valgrind, uProf, Logic Analyser
- **Distributed**: Apache Spark, Hadoop, Kafka, JVM internals, Multithreading, Concurrency, Microservices
- **Cloud & Infra**: AWS, Microsoft Fabric, Azure Data Factory, Docker, Linux, Git, CI/CD

Open Source Contributions (12 merged PRs across Meta Velox, PyTorch, Apache Spark & Apache Gluten)

- **Meta Velox**: facebookincubator/velox – **5 merged**: Parquet WriterOptions encoding, geospatial docs (`convex_hull_agg`, `geometry_union_agg`), Spark `transform_values` registration, TPC-DS map allocation optimization, Sphinx doc fix.
- **Apache Spark**: apache/spark – **3 merged**: `RuntimeConfig.get` docs; JSON `inferTimestamp` option; `singleVariantColumn` for CSV/JSON/XML. *Open*: streaming `GroupState`, Protobuf errors.
- **PyTorch**: pytorch/pytorch – **2 merged**: schema-inference error messaging; `ValueError` for `norm_type=0` in `LPPoolNd`. *Open*: `fp16` CPU/GPU parity, CUDA device-guard RAII.
- **Apache Gluten**: apache/gluten – **2 merged**: re-enabled `atan2` test; compression-codec error message in native shuffle. *Open*: AQE bloom-filter corruption fix.
- **vLLM**: vllm-project/vllm – *Open*: gate `Step3VL` multimodal test under Transformers v5.

Achievements

Google Code Jam 2020 – Global Rank **6,448 / 44,434** · GATE CSE 2019 – **97th Percentile** · TCS CodeVita 2016 – **414 AIR** (1st college rank)